

EXACT DISTRIBUTIONS OF THE PRODUCT OF CERTAIN INDEPENDENT MATRICES

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Key words: Exact distribution, matrix product, Wishart, Beta, Mellin transform, hypergeometric function, G-function, Bessel function

1. Introduction

Let X be a random $p \times p$ positive definite matrix whose probability density function (p.d.f.) is given by

$$(1.1) \quad f(X) = \begin{cases} C|X|^{n-\frac{1}{2}(p+1)} {}_1F_1(\alpha; \theta; -X), & X > 0 \\ 0 & \text{otherwise} \end{cases}$$

where ${}_1F_1(\alpha; \theta; -X)$ is a confluent hypergeometric function with matrix argument (Constantine (1963)) and

$$C = \frac{\Gamma_p(\alpha)\Gamma_p(\theta-n)}{\Gamma_p(n)\Gamma_p(\theta)\Gamma_p(\alpha-n)}$$

with $\text{Re}(\theta-n) > \frac{1}{2}(p-1)$, $\text{Re}(\alpha-n) > \frac{1}{2}(p-1)$ and the parameters α and θ are restricted to take values for which $f(X)$ is positive.

Form (1.1) was introduced by Van der Merwe and Roux (1974). It was shown by them that the familiar Beta distribution of the second kind as well as the Wishart distribution are special cases.

In this paper the exact distribution of the product of two independent matrices taken from samples with p.d.f. given by

- (i) (1.1) above
- (ii) a multivariate beta type II, and
- (iii) a Wishart

distribution is given.

Distributions of the products of two independent random determinants are also investigated for the above three cases using a technique based on their h -th order moments and the Mellin's inversion theorem. The results arrived at are in the most general form i.e., in terms of Meijer G-functions.

Before we actually proceed to obtain the distributions in Sections 3 and 4, we list, in Section 2, some useful results and definite integrals taken from James (1964), Constantine (1963, 1966), Khatri (1966) and Herz (1955).

2. Notations and preliminary results

In the following section, use will be made of zonal polynomials and hypergeometric functions of matrix argument. For convenience we shall list a few lemmas below. Definitions, proofs, etc. may be found in the papers listed above.

Lemma 2.1 (Khatri (1966))

Let Z be a complex symmetric matrix such that $R(Z) > 0$ and let T be an arbitrary complex symmetric matrix. Then

$$(2.1) \int_{S > 0} e^{-\text{tr} ZS} |S|^{t-\frac{1}{2}(p+1)} C_{\kappa}(TS^{-1}) dS = \Gamma_p(t, -\kappa) |Z|^{-t} C_{\kappa}(ZT),$$

where $R(t) > \frac{1}{2}(p-1) + k_1$.

Lemma 2.2 (Khatri (1966))

If R is any positive definite $p \times p$ matrix then

$$(2.2) \int_0^1 |S|^{t-\frac{1}{2}(p+1)} |I-S|^{u-\frac{1}{2}(p+1)} C_{\kappa}(RS) dS = \frac{\Gamma_p(t, \kappa) \Gamma_p(u) C_{\kappa}(R)}{\Gamma_p(t+u, \kappa)}.$$

Lemma 2.3 (Constantine (1966))

The Laplace transform of the zonal polynomial is for $R(a) > k_1 + \frac{1}{2}(p-1)$

$$(2.3) \int_{R > 0} e^{-\text{tr} RZ} |R|^{a-\frac{1}{2}(p+1)} C_{\kappa}(R^{-1}) dR = \frac{(-1)^k \Gamma_p(a)}{(-a + \frac{1}{2}(p+1))_{\kappa}} |Z|^{-a} C_{\kappa}(Z).$$

Lemma 2.4 (Khatri (1966))

If R is any arbitrary symmetric complex $p \times p$ matrix then

$$(2.4) \int_{S > 0} |S|^{t-\frac{1}{2}(p+1)} |I+S|^{-t-u} C_{\kappa}(RS^{-1}) dS = \frac{\Gamma_p(t, -\kappa) \Gamma_p(u, \kappa)}{\Gamma_p(t+u)} C_{\kappa}(R).$$

Lemma 2.5 (Khatri (1966))

Let R be a positive definite matrix. Then for $t \geq \frac{1}{2}(p+k_1)$

$$(2.5) \int_0^1 |S|^{t-\frac{1}{2}(p+1)} |I-S|^{u-\frac{1}{2}(p+1)} C_{\kappa}(RS^{-1}) dS = \frac{\Gamma_p(t, -\kappa) \Gamma_p(u)}{\Gamma_p(t+u, -\kappa)} C_{\kappa}(R).$$

From (2.1) and (2.3) it thus follows that

$$(2.6) \quad \Gamma_p(t, -\kappa) = \frac{(-1)^k \Gamma_p(t)}{(-t + \frac{1}{2}(p+1))_\kappa}$$

We shall further make use of the integral representation of the hypergeometric function namely of ${}_1F_1$ as given by Herz (1955) :

$$(2.7) \quad {}_1F_1(\beta; \theta; X) = \frac{\Gamma_p(\theta)}{\Gamma_p(\beta) \Gamma_p(\theta - \beta)} \int_0^1 e^{\text{tr} X T} |T|^{\beta - \frac{1}{2}(p+1)} |I - T|^{\theta - \beta - \frac{1}{2}(p+1)} dT$$

as well as the definition (Constantine (1963))

$$(2.8) \quad {}_pF_q(\beta_1, \dots, \beta_p; \theta_1, \dots, \theta_q; S) = \sum_{k=0}^{\infty} \sum_{\kappa} \frac{(\beta_1)_{\kappa} \dots (\beta_p)_{\kappa}}{(\theta_1)_{\kappa} \dots (\theta_q)_{\kappa}} \frac{C_{\kappa}(S)}{k!}$$

and as special cases:

$$(2.9) \quad {}_0F_0(X) = e^{\text{tr} X} \quad \text{and}$$

$$(2.10) \quad {}_1F_0(\alpha; X) = |I - X|^{-\alpha}.$$

3. Exact distributions of the products of certain independently distributed matrices

In this section we shall derive the exact distribution of certain independently distributed matrices. To obtain these use was made of the results given in Section 2.

Theorem 3.1

Let $X_i; p \times p, i=1,2$ be random matrices independently distributed with the following density functions (Van der Merwe and Roux (1974)):

$$(3.1) \quad f_i(X_i) = \begin{cases} C_i |X_i|^{n_i - \frac{1}{2}(p+1)} {}_1F_1(\beta_i; \theta_i; -X_i), & X_i > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $i=1,2$ and

$$C_i = \frac{\Gamma_p(\beta_i) \Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\theta_i) \Gamma_p(\beta_i - n_i)}$$

respectively. Then $U = X_2^{1/2} X_1 X_2^{1/2}$ has the density,

$$(3.2) \quad g_1(U) = \begin{cases} \frac{\Gamma_p(\beta_1) \Gamma_p(\theta_1 - n_1) \Gamma_p(\theta_2 - n_2) \Gamma_p(n_2 - n_1) \Gamma_p(n_1 - n_2 + \beta_2)}{\Gamma_p(n_1) \Gamma_p(\theta_1) \Gamma_p(\beta_1 - n_1) \Gamma_p(n_2) \Gamma_p(\beta_2 - n_2) \Gamma_p(n_1 - n_2 + \theta_2)} \\ \cdot |U|^{n_1 - 1/2(p+1)} {}_2F_3(\beta_1, n_1 - n_2 + \beta_2; n_1 - n_2 + 1/2(p+1), \theta_1, \\ n_1 - n_2 + \theta_2; U), & U > 0 \\ 0 & \text{elsewhere} \end{cases}$$

provided $R(n_2 - n_1) > 1/2(p-1) + k_1$.

Proof

In the joint density of X_1 and X_2 put $U = X_2^{1/2} X_1 X_2^{1/2}$ where $J(U \rightarrow X_1) = |X_2|^{1/2(p+1)}$ (Deemer and Olkin (1951)) then

$$g_1(U) = C_1 C_2 |U|^{n_1 - 1/2(p+1)} \int_{X_2 > 0} |X_2|^{n_2 - n_1 - 1/2(p+1)} {}_1F_1(\beta_1; \theta_1; -U X_2^{-1}) {}_1F_1(\beta_2; \theta_2; -X_2) dX_2.$$

Using (2.7) we obtain

$$(3.3) \quad g_1(U) = C_1 C_2 \frac{\Gamma_p(\theta_1) \Gamma_p(\theta_2)}{\Gamma_p(\beta_1) \Gamma_p(\theta_1 - \beta_1) \Gamma_p(\beta_2) \Gamma_p(\theta_2 - \beta_2)} I_1$$

$$\begin{aligned} \text{where } I_1 &= \int_{X_2 > 0} |X_2|^{n_2 - n_1 - 1/2(p+1)} |U|^{n_1 - 1/2(p+1)} \\ &\cdot \int_0^1 e^{-\text{tr} R U X_2^{-1}} |R|^{\beta_1 - 1/2(p+1)} |I - R|^{\theta_1 - \beta_1 - 1/2(p+1)} \\ &\cdot \int_0^1 e^{-\text{tr} S X_2} |S|^{\beta_2 - 1/2(p+1)} |I - S|^{\theta_2 - \beta_2 - 1/2(p+1)} dR dS dX_2. \end{aligned}$$

Changing the order of integration in I_1 and using (2.9) and lemma 2.1 we obtain

$$\begin{aligned}
I_1 &= \sum_{k=0}^{\infty} \sum_{\kappa} \frac{1}{k!} \Gamma_p(n_2 - n_1, -\kappa) \\
&\cdot \int_0^1 \int_0^1 |R|^{\beta_1 - \frac{1}{2}(p+1)} |I-R|^{\theta_1 - \beta_1 - \frac{1}{2}(p+1)} |U|^{n_1 - \frac{1}{2}(p+1)} \\
&\cdot |S|^{n_1 - n_2 + \beta_2 - \frac{1}{2}(p+1)} |I-S|^{\theta_2 - \beta_2 - \frac{1}{2}(p+1)} C_{\kappa}(-SRU) dR dS.
\end{aligned}$$

Integrating over S and then over R using lemma 2.2 (Khatri 1966) together with (2.6) and (2.8) we obtain

$$\begin{aligned}
I_1 &= \frac{\Gamma_p(\beta_1) \Gamma_p(\theta_2 - \beta_2) \Gamma_p(n_1 - n_2 + \beta_2) \Gamma_p(\theta_1 - \beta_1) \Gamma_p(n_2 - n_1)}{\Gamma_p(\theta_1) \Gamma_p(n_1 - n_2 + \theta_2)} \\
&\cdot {}_2F_3(\beta_1, n_1 - n_2 + \beta_2; \beta_1, n_1 - n_2 + \theta_2, n_1 - n_2 + \frac{1}{2}(p+1); U).
\end{aligned}$$

Using this expression in (3.3) the theorem follows.

Theorem 3.2

Let $X_i: p \times p$, $i=1,2$ be random matrices independently distributed with the following Beta II (Van der Merwe and Roux (1974)), density functions

$$(3.4) \quad f_2(X_i) = \begin{cases} D_i |X_i|^{n_i - \frac{1}{2}(p+1)} {}_1F_0(\beta_i; -X_i), & X_i > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $i=1,2$ and $D_i = \frac{\Gamma_p(\beta_i)}{\Gamma_p(n_i) \Gamma_p(\beta_i - n_i)}$ respectively.

Then $U = X_2^{\frac{1}{2}} X_1 X_2^{\frac{1}{2}}$ has the density:

$$(3.5) \quad g_2(U) = \begin{cases} \frac{\Gamma_p(\beta_1) \Gamma_p(n_2 - n_1) \Gamma_p(\beta_2 + n_1 - n_2)}{\Gamma_p(n_1) \Gamma_p(n_2) \Gamma_p(\beta_1 - n_1) \Gamma_p(\beta_2 - n_2)} \\ \cdot |U|^{n_1 - \frac{1}{2}(p+1)} {}_2F_1(\beta_1, \beta_2 + n_1 - n_2; n_1 - n_2 + \frac{1}{2}(p+1); U), & U > 0 \\ 0 & \text{elsewhere} \end{cases}$$

provided $R(n_2 - n_1) \geq \frac{1}{2}(p + k_1)$.

Proof

In the joint density of X_1 and X_2 put $U = X_2^{\frac{1}{2}} X_1 X_2^{\frac{1}{2}}$ where $J(U, X_1) = |X_2|^{\frac{1}{2}(p+1)}$ then from (2.10) we have

$$g_2(U) = D_1 D_2 |U|^{n_1 - \frac{1}{2}(p+1)} \\ \cdot \int_{X_2 > 0} |X_2|^{n_2 - n_1 - \frac{1}{2}(p+1)} |I + X_2|^{-\beta_2} \\ \cdot {}_1F_0(\beta_1; -X_2^{-1}U) dX_2.$$

Using (2.8), (2.4) and (2.6) the theorem is proved.

Theorem 3.4

Let $X_i: p \times p$, $i=1,2$ be random matrices independently distributed with the following, Wishart, density functions

$$(3.6) f_3(X_i) = \begin{cases} G_i |X_i|^{n_i - \frac{1}{2}(p+1)} {}_0F_0(-X_i), & X_i > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $i=1,2$ and

$$G_i = \frac{1}{\Gamma_p(n_i)}$$

respectively. Then $U = X_2^{\frac{1}{2}} X_1 X_2^{\frac{1}{2}}$ has the density

$$g_3(U) = \begin{cases} \frac{\Gamma_p(n_2 - n_1)}{\Gamma_p(n_1) \Gamma_p(n_2)} |U|^{n_1 - \frac{1}{2}(p+1)} {}_0F_1(n_1 - n_2 + \frac{1}{2}(p+1); U), & U > 0 \\ 0 & \text{elsewhere} \end{cases}$$

provided $R(n_2 - n_1) > \frac{1}{2}(p-1) + k_1$.

Proof

The proof follows from Theorem 3.1 by putting for $i=1,2$ $\beta_i = \theta_i$.

By noting, Herz (1955), that

$$B_{\delta}(Z) = \int_{\Lambda > 0} e^{-\text{tr} \Lambda Z} e^{-\text{tr} \Lambda^{-1}} |\Lambda|^{\delta - \frac{1}{2}(p+1)} d\Lambda$$

where $B_{\delta}(Z)$ is called the Bessel function of the second kind of order δ ;

writing

$$K_{\delta}(T) = \frac{1}{2^p} B_{\delta}(\frac{1}{2} T' T)^{\frac{1}{2}} T^{\delta}$$

$$\text{with } K_{\delta}(Z) = \frac{1}{2^p} \int_{R > 0} e^{-\frac{1}{2} \text{tr}(R + R^{-1})Z} |R|^{\delta - \frac{1}{2}(p+1)} dR$$

and where $K_{\delta}(T)$ is the modified Bessel function of the second kind of order δ ; we may write

$$K_{n_2 - n_1}(2U^{\frac{1}{2}}) = \Gamma_p(n_2 - n_1) |U|^{-\frac{1}{2}(n_2 - n_1)} {}_0F_1(n_1 - n_2 + \frac{1}{2}(p+1); U)$$

where $U = U' U^{\frac{1}{2}}$.

Thus

$$g_3(U) = \frac{2^p}{\Gamma_p(n_1) \Gamma_p(n_2)} |U|^{\frac{1}{2}(n_1 + n_2) - \frac{1}{2}(p+1)} K_{n_2 - n_1}(2U^{\frac{1}{2}}), U > 0.$$

This is a direct generalisation to matrices of a result due to Malik (1968) for the univariate case, if we consider the Wishart distribution as a matrix extension of the univariate gamma distribution.

4. Distributions of products of two independent random determinants

Here we shall consider the class of probability functions defined by (1.1) and we will study the distributions of the products of independent random determinants whose probability laws belong to this class.

Theorem 4.1

Let X_1, X_2 be two independent random matrices having the probability laws as given in (3.1) with $\alpha_i = \beta_i$, where $i=1,2$ and the parameters have the same meanings as before, then $v = |X_1| |X_2|$ has the probability function,

$$(4.1) \quad h_I(v) = \begin{cases} \left[\frac{2}{\prod_{i=1}^{2p} \frac{\Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i)}} \right] \pi^{\frac{1}{2}p(p-1)} \\ G_{2p}^{2p} \quad \begin{matrix} 2p & 2p \\ 4p & 4p \end{matrix} \quad \left[v \mid \begin{matrix} a_1, \dots, a_{2p} \\ b_1, \dots, b_{4p} \end{matrix} \right] & \text{for } v > 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{where } a_i = \begin{cases} n_1 - \alpha_1 + \frac{1}{2}(i-1) & i=1, 3, \dots, 2p-1 \\ n_2 - \alpha_2 + \frac{1}{2}(i-2) & i=2, 4, \dots, 2p \end{cases}$$

$$\text{and } b_j = \begin{cases} n_1 - \frac{1}{2}(j+3) & j=1, 3, \dots, 2p-1 \\ n_2 - \frac{1}{2}(j+2) & j=2, 4, \dots, 2p \\ n_1 - \theta_1 + \frac{1}{2}(j-2p-1) & j=2p+1, 2p+3, \dots, 4p-1 \\ n_2 - \theta_2 + \frac{1}{2}(j-2p-2) & j=2p+2, 2p+4, \dots, 4p \end{cases}$$

and the G function is defined by (Mathai (1971))

$$(4.2) \quad G_{p \quad q}^{r \quad s} \left[x \mid \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix} \right] = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} \frac{\prod_{j=1}^r \Gamma(b_j + h) \prod_{j=1}^s \Gamma(1 - a_j - h)}{\prod_{j=r+1}^q \Gamma(1 - b_j - h) \prod_{j=s+1}^p \Gamma(a_j + h)} x^{-h} dh$$

Proof

Consider the h -th moment of v , namely

$$(4.3) \quad E(v^h) = \int_{X_1 > 0} |X_1|^h f(X_1) dX_1 \int_{X_2 > 0} |X_2|^h f(X_2) dX_2.$$

By using the fact that $\int_{X_i > 0} f(X_i) dX_i = 1$

it follows from (3.1) and (4.3) that

$$(4.4) \quad E(v^h) = \left[\frac{2}{\prod_{i=1}^{2p} \frac{\Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i)}} \right] \left[\frac{\Gamma_p(n_1 + h) \Gamma_p(n_2 + h) \Gamma_p(\alpha_1 - n_1 - h) \Gamma_p(\alpha_2 - n_2 - h)}{\Gamma_p(\theta_1 - n_1 - h) \Gamma_p(\theta_2 - n_2 - h)} \right]$$

where the real parts in $\Gamma(\cdot)$ are positive. After comparing the multivariate gamma functions in the square brackets in (4.4) it follows that

$$E(v^h) = \left[\prod_{i=1}^2 \frac{\Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i)} \right] \pi^{1/2} p(p-1) \cdot \frac{\left[\prod_{j=1}^{2p} \Gamma(c_j + h) \prod_{j=1}^{2p} \Gamma(1 - d_j - h) \right]}{\left[\prod_{j=2p+1}^{4p} \Gamma(d_j + h) \prod_{j=2p+1}^{4p} \Gamma(1 - c_j - h) \right]}$$

$$\text{where } c_j = \begin{cases} n_1 - \frac{1}{2}(j-1) & j=1, 3, 5, \dots, 2p-1 \\ n_2 - \frac{1}{2}(j-2) & j=2, 4, \dots, 2p \\ 1 - \theta_1 + n_1 + \frac{1}{2}(j-2p-1) & j=2p+1, 2p+3, \dots, 4p-1 \\ 1 - \theta_2 + n_2 + \frac{1}{2}(j-2p-2) & j=2p+2, 2p+4, \dots, 4p \end{cases}$$

$$\text{and } d_j = \begin{cases} 1 - \alpha_1 + n_1 + \frac{1}{2}(j-1) & j=1, 3, 5, \dots, 2p-1 \\ 1 - \alpha_2 + n_2 + \frac{1}{2}(j-2) & j=2, 4, \dots, 2p. \end{cases}$$

By considering the inverse Mellin transform (Erdelyi *et al* (1954)),

$$(4.5) \quad f(v) = (2\pi i)^{-1} \int_{c-i\infty}^{c+i\infty} E(v^h) v^{-h-1} dh$$

we obtain the probability function of the random variable v as given by

$$h_1(v) = \left[\prod_{i=1}^2 \frac{\Gamma_p(\theta_i - n_i)}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i)} \right] \pi^{1/2} p(p-1) \\ v^{-1} G_{2p}^{2p} \left[\begin{matrix} d_1, \dots, d_{2p} \\ c_1, \dots, c_{4p} \end{matrix} \middle| v \right].$$

Using Erdelyi (1953) 5.3(8) the theorem is proved.

Theorem 4.2

Let X_1 and X_2 be two independent random matrices with density functions as given by (3.4) with $\alpha_i = \beta_i$, for $i=1, 2$ respectively. Then $w = |X_1| |X_2|$ has the density

$$(4.6) \quad h_2(w) = \begin{cases} \prod_{i=1}^2 \left\{ \frac{1}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i)} \right\} \cdot \pi^{p(p-1)/2} w^{-1} \\ \cdot G_{2p}^{2p} \left[\begin{matrix} a_1, \dots, a_{2p} \\ b_1, \dots, b_{2p} \end{matrix} \middle| w \right] & \text{for } w > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $b_i = \begin{cases} n_1 - \frac{1}{4}(i-1) & i=1, 3, \dots, 2p-1 \\ n_2 - \frac{1}{4}(i-2) & i=2, 4, \dots, 2p \end{cases}$

$a_i = \begin{cases} 1 - \alpha_1 + n_1 + \frac{1}{4}(i-1) & i=1, 3, \dots, 2p-1 \\ 1 - \alpha_2 + n_2 + \frac{1}{4}(i-2) & i=2, 4, \dots, 2p \end{cases}$

and the G function is defined by (4.2).

Proof

Using the fact that $\int_{X_i > 0} f_2(X_i) dX_i = 1$

we may write

$$E(w^h) = \prod_{i=1}^2 \left\{ \frac{1}{\Gamma_p(n_i) \Gamma_p(\alpha_i - n_i)} \right\} \\ [\Gamma_p(n_1 + h) \Gamma_p(n_2 + h) \Gamma_p(\alpha_1 - n_1 - h) \Gamma_p(\alpha_2 - n_2 - h)] .$$

Comparing the multivariate gamma functions in the square brackets and using (4.5) the theorem is proved.

For $p=1$ and $k=2$ this result reduces to the result in Oliver (1972) paragraph 7.

Theorem 4.3

Let X_1, X_2 be two independent random matrices having the probability laws $f_2(X_i)$ for $i=1, 2$ respectively, then

$u = |X_1| |X_2|$ has the probability function

$$h_3(u) = \begin{cases} \frac{1}{\Gamma_p(n_1)\Gamma_p(n_2)} \pi^{\frac{1}{2}p(p-1)} u^{-1} \\ G_0^{2p} \begin{matrix} 0 \\ 2p \end{matrix} (u|a_1, \dots, a_{2p}) \text{ for } u > 0 \\ 0 \text{ elsewhere} \end{cases}$$

$$\text{with } a_i = \begin{cases} n_1 - \frac{1}{2}(i-1) & i=1, 3, \dots, 2p-1 \\ n_2 - \frac{1}{2}(i-2) & i=2, 4, \dots, 2p \end{cases}$$

and the G function defined by (4.2).

Proof

From (4.4) with $\theta_i = \alpha_i$ $i=1, 2$ we have

$$E(u^h) = \frac{1}{\Gamma_p(n_1)\Gamma_p(n_2)} \{ \Gamma_p(n_1 + h) \Gamma_p(n_2 + h) \}.$$

Now, using (4.5) we can obtain the density function $h_3(u)$ of u from (4.4) as given by the theorem.

We note that for $p=1$ this result reduces to that of Malik (1968) because (Erdelyi *et al* (1953) 5.6(4))

$$G_0^{2 \quad 0} \begin{matrix} 0 \\ 2 \end{matrix} (u|n_1, n_2) = 2u^{\frac{1}{2}(n_1+n_2)} K_{n_1-n_2}(2u^{\frac{1}{2}}).$$

The author wishes to thank Dr J.J.J. Roux for his helpful suggestions and discussions.

REFERENCES

- CONSTANTINE, A.G. (1963). Some non-central distribution problems in multivariate analysis. *Ann. Math. Statist.* **34**, 1270-1285.
- CONSTANTINE, A.G. (1966). The distribution of Hotelling's generalised T_0^2 . *Ann. Math. Statist.*, **37**, 215-225.
- DEEMER, W.L. and OLKIN, I. (1951). The Jacobians of certain matrix transformations useful in multivariate analysis. *Biometrika*, **38**, 345-367.

- ERDÉLYI, A. *et al.* (1953). *Higher Transcendental Functions. Vol. 1*, McGraw-Hill New York.
- ERDÉLYI, A. *et al.* (1954). *Tables of integral transforms. Vol. 1*, McGraw-Hill New York.
- HERZ, C.S. (1955). Bessel function of matrix argument. *Ann. Math.*, **61**, 474-523.
- JAMES, A.T. (1964). Distribution of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.*, **35**, 475-501.
- KHATRI, C.G. (1966). On certain distribution problems based on positive definite quadratic functions in normal vectors. *Ann. Math. Statist.*, **37**, 468-479.
- MALIK, H.J. (1968). Exact distribution of the product of independent generalised Gamma variables with the same parameters. *Ann. Math. Statist.*, **39**, 1751-1752.
- MATHAI, A.M. (1971). An expansion of Meyer's G-function and the distribution of products of independent beta variates. *S. Afr. Statist. J.*, **5**, 71-90.
- OLIVER, P.C. Du V. (1972). Contributions to Normal Sampling Theory. *S. Afr. Statist. J.*, **6**, 53-82.
- VAN DER MERWE, G.J. and ROUX, J.J.J. (1974). On a generalised matrix-variate hypergeometric distribution. *S. Afr. Statist. J.*, **8**, 49-58.

Manuscript received 27 November 1973